

Video Processing and Optimisation

Shashwat Saxena

July 16, 2026

1 Introduction

1.1 Problem Statement

Conventional methods for video compression are based on slicing frames and often lead to temporal discontinuity and poor results in video with motion. This research aims to develop a motion-compensated temporal frame fusion algorithm that reduces frames by integrating information from multiple consecutive frames onto one net frame, thereby preserving information and allowing data reduction.

2 Mathematical Background

Video could be thought of a 3D signal, with the domain being time ($\mathbf{R}$) and the co-domain being images ($\mathbf{R}$). At each instance of time, an image is being displayed, an image itself is a signal that maps a 2D space($\mathbf{R}^2$) onto light intensity of 3 colors (RGB), ($\mathbf{R} \times \mathbf{R} \times \mathbf{R}$)

2.1 Block Averaging

The most simple method to deal with this problem would be to algebraically add and take an average of all the frames in the group, this would result in a reduced framerate but, wouldn't be ideal as it wouldn't compensate for the motion of different objects.

Simple algebraic addition leads to loss of key details in the environment as well which would be important for the depth perception RL model (monocular depth measuring).

2.2 Integration

A problem with actual integration is that, unlike real world video, which is a continuous time 3D signal, video stored digitally is broken up into frames, hence performing integration isn't possible

Let I be a 3D signal representing a video,

$$I(x, y, t) : \mathbf{R}^2 \times \mathbf{R} \rightarrow \mathbf{R}$$

To average the first one second,

$$\int_0^1 I(x, y, t) dt$$

As this integration is not possible with discrete time domain of digital videos, we must convert the integral into its equivalent Reiman Sum (Discrete Reimann Sum),

$$\sum_0^1 I(x, y, t) \cdot \Delta t$$

$$\Delta t = \frac{1}{\text{fps}}$$

Since in most cases $\int_0^\infty x^2 dx$

3 Notes